EMBER — fire intelligence at the first flicker

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Team

Field	Answer
Team name	[to fill]
Property	3 — Fire Detection
Members	[Member A], [Member B], [Member C]
Team leader	[to fill]
Constraint card	[write the card's rule here + one line on how EMBER follows it]

Member A = hardware and firmware. Member B = AI and data. Member C = dashboard, security and pitch. With 2 members, B and C merge.

EMBER in simple words

EMBER is a room detector that spots the early signs of a fire, tells a real danger apart from false alarms, sounds a clear alarm and closes the gas before things get worse.

1. **It watches three signs at once:** gas or smoke in the air, heat that rises fast, and the flicker of a flame.
2. **A small AI model decides.** Every 0.5 s it reads the last 5 seconds of readings and gives a danger level: NORMAL, WARNING or ALARM. Gas alone means a leak (WARNING). Gas + fast heat + flicker means fire (ALARM).
3. **It is not fooled.** A flashlight is bright but steady. A warm hand is heat with no flame. One odd reading never starts the alarm.
4. **It acts on its own, with no internet:** green, yellow or red LED, a different siren for each level, and the gas valve closes.
5. **It tells people.** The dashboard shows live values, the level, why it decided, and the event history.

6. **Only an allowed person can reset it,** from the dashboard with a login. Every command is signed, so a fake command is rejected.
7. **It keeps working when a part fails.** A broken sensor is reported as a fault. If the connection drops, the board keeps protecting the room on its own.

The problem in our own words

Who is affected: families in homes that cook with butane bottles, small restaurants and workshops, and the building managers responsible for them.

What can fail: a gas leak goes unnoticed and turns into an explosion or a fire. A small fire starts at night or in an empty room and nobody sees it in time. Classic detectors give false alarms, so people ignore them or unplug them. A detector that only beeps locally helps nobody when the room is empty.

What it costs: lives and injuries first, then the home or business, its stock and its income. Every minute of early warning gives people more time to leave and stops a small fire from becoming a big one.

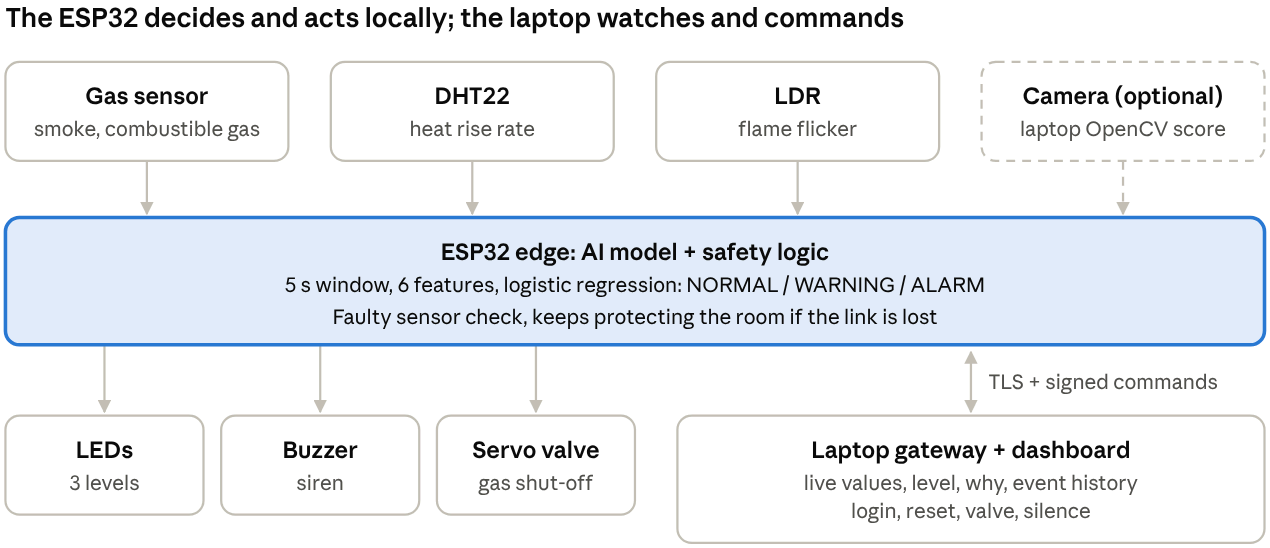
Our six layers

Layer	What we will do	Who is in charge
Sensing	Three sensors of different kinds: gas sensor (smoke and combustible gas), DHT22 (temperature, used for how fast heat rises), LDR sampled fast (flame flicker). Plausibility checks flag a faulty or disconnected sensor: DHT22 returns no value, gas reading stuck at 0 or max, LDR flat. One odd reading never starts an alarm: decisions use a 5 s window.	Member A
Communication	The ESP32 sends a reading packet every 0.5 s over Wi-Fi to the laptop gateway, and receives commands back. Traffic is protected with HTTPS (TLS, pinned certificate). Every command is also signed with HMAC-SHA256 plus a counter, so a copied or replayed command is rejected.	Member C

Layer	What we will do	Who is in charge
Edge processing and AI	A logistic regression model runs on the ESP32. It reads a 5 s window (10 readings) and computes 6 features: gas above baseline, gas slope, heat slope, light change, flicker, camera flame score (optional, laptop OpenCV). Output: probability of normal, gas leak, fire, mapped to NORMAL / WARNING / ALARM. Trained on data we record, including traps (flashlight, warm hand). Tested on at least 20 labeled windows with a confusion matrix, accuracy, precision and recall. Key metric: fire recall. Compared to a simple threshold rule.	Member B
Actuation	Automatic: LEDs green, yellow, red; buzzer with one pattern per level; servo as the gas shut-off valve, closed at gas leak or fire. Manual: an allowed user can open or close the valve and silence the buzzer from the dashboard. Safe state on failure or lost link: valve closed, local alarm keeps running.	Member A
Security	Threat 1: someone resets or silences the alarm without the right to (prank, intruder). Protection: dashboard login, reset needs an authorized account, the board's button can only silence for 60 s, never reset. Threat 2: fake or replayed commands on the network (for example "open valve"). Protection: HMAC-SHA256 signature + counter, rejected and logged. Threat 3: eavesdropping or tampered data in transit. Protection: TLS. Threat 4: connection lost or jammed. Protection: the board decides and acts alone.	Member C

Layer	What we will do	Who is in charge
Operator interface	Web dashboard on the laptop: live values, a big colored status banner (NORMAL / WARNING / ALARM), fire probability with "why" bars, sensor fault indicators, event history, commands (authorized reset, silence, valve open or close). A non-expert reads it in a few seconds.	Member C

Our architecture



EMBER architecture · sensing, edge AI, actuation, gateway

The board senses, decides and reacts without the laptop; the laptop adds the camera score, the dashboard and authorized commands.

Target	Where it is handled
Three danger levels	Model output on the ESP32, shown on the LEDs and the dashboard banner
Alarm within 20 s	0.5 s loop on a 5 s window: alarm in under 10 s
No alarm in normal conditions	Model trained on traps (flashlight, warm hand) + 5 s window against odd readings

Target	Where it is handled
Only an allowed person resets	Dashboard login + signed reset command; the board button only silences for 60 s

Parts list

Total: 475 of 490 credits. Three sensors of different kinds, three kinds of actuators, one board.

Part	Role	Credits	Second choice if sold out
ESP32 development board	Brain, Wi-Fi, runs the model	150	Arduino Uno (100), model runs on the laptop over USB
Gas sensor	Smoke and combustible gas	90	None in the bank: keep DHT22 + LDR, buy the second-choice relay with the saved credits
DHT22	Temperature, heat rise rate	70	None in the bank: gas + LDR still give two sensors
LDR light sensor	Flame flicker	30	Infrared obstacle sensor (40)
Servo motor	Gas shut-off valve	70	Small fan (80) as a vent, gas leak only
Buzzer	Siren, one pattern per level	20	LEDs only
LED red, yellow, green	Danger level on the board	30	Fewer LEDs + level shown in the buzzer pattern
Push button	Local silence for 60 s, never a reset	10	Dashboard only
Extra wires, male to female	Spare wiring	5	Starter pack only

Free starter pack: breadboard, 30 jumper wires, resistors, USB power. The laptop webcam is ours, so the camera layer costs nothing.

The real-life solution

In a real building, EMBER is one detector per room linked to a building gateway, a manager dashboard and the emergency services.

- **Multi-sensor detector in each room:** certified gas and smoke sensor, heat sensor, infrared flame sensor, optional camera for visual confirmation.
- **AI on the detector:** the same windowed model gives three levels and separates a gas leak from a fire. It keeps working without internet.
- **Automatic reaction:** loud siren and strobe, gas solenoid valve that closes, ventilation shut down during a fire, power to the room cut.
- **Alerts:** building manager notified at every level change. Pre-alert to Civil Protection (198) at WARNING, dispatch at ALARM, with room, floor, intensity and the gas hazard.
- **Multi-room dashboard:** every room on one screen, event history, sensor health.
- **Authorized reset only:** user accounts with roles, signed commands, encrypted traffic.
- **Reliability:** battery backup, sensor self-check reported as a fault, regular automatic tests, certification before sale (EN 54 family in Europe).

How our prototype maps to the real solution

The prototype keeps the full chain (sense, decide, react, alert, authorized reset) and only swaps parts for cheaper, safer ones.

What changes

Prototype	Real solution	Why the change
LDR reading flicker	Infrared flame sensor	Same signal (flicker), certified and immune to room light
Hobby gas sensor	Certified gas and smoke sensor	Precision, long-term stability, certification
Servo turning a model valve	Gas solenoid valve, closed without power	Real shut-off, fail-safe on power loss
Flame video and unlit gas as test inputs	Real smoke, heat and flame	No real fire is allowed at the event
Laptop as gateway and dashboard	Building gateway + cloud dashboard	Runs 24/7, many rooms

Prototype	Real solution	Why the change
Local Wi-Fi hotspot	Building Wi-Fi or LoRa, TLS	Range and reliability
One room, USB power	One detector per room, mains + battery	Coverage and power cuts

What is preserved

- The same three early signs: gas or smoke, heat rise rate, flame flicker.
- The same AI: a model on the detector reading a 5 s window, three levels, leak vs fire.
- The same safety logic: acts without internet, safe state on failure, faulty sensor reported.
- The same security: login, signed commands, encrypted traffic, authorized reset.
- The same dashboard: live values, level, why, history, commands.

Why the prototype stays representative: the decision logic, the data flow and the security model are identical. Only the sensors and the valve get better versions with the same interface, so the code and the model carry over after retraining on real sensor data.

Estimated unit cost of the real deployment (approximate, to verify): about 100 to 160 DT per room detector, plus about 60 to 100 DT per gas valve, plus one gateway per building. Certification and installation come on top.